

Prakhar Kumar

Full Stack Developer | Backend Developer | Software Engineer

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience delivering **2+ production-grade applications** using **React.js, Node.js, TypeScript, and REST APIs**. Proficient in designing scalable backend systems, automating workflows, optimizing relational and NoSQL database schemas, and ensuring software quality through structured testing. Achieved a **40% improvement in ticket resolution efficiency**, a **25% enhancement in data accuracy**, and a **sub-200ms WebSocket latency** under concurrent load. IEEE-published researcher with proven experience in Agile/Scrum development environments.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication, WebSocket (Socket.IO)

Databases: MongoDB, MySQL

Tools: Git, GitHub, Postman, Linux

Concepts: OOP, DBMS, Data Structures and Algorithms, MVC Architecture, RBAC, SDLC, Agile/Scrum

EDUCATION

Chandigarh University, Mohali, Punjab

Bachelor of Engineering (B.E.) – Computer Science and Engineering

2022 – 2026

CGPA: 8.05 / 10.0

PROJECTS

SupportSync – AI-Powered Real-Time Customer Support Platform | [Live](#) | [GitHub](#)

2026

- Architected a scalable, full-stack real-time support system using **Node.js, React.js, TypeScript, and Socket.IO**; resolved event-driven concurrency issues through structured log analysis, sustaining **sub-200ms WebSocket latency** under high concurrent load.
- Designed **automated AI-to-human ticket escalation workflows** secured by **Role-Based Access Control (RBAC)** and **JWT authentication**; executed end-to-end **User Acceptance Testing (UAT)**, reducing ticket resolution time by **40%**.
- Built and validated **RESTful APIs** with **Express.js** following **MVC architecture** and **OOP** principles; optimized **MongoDB** schemas for efficient session and chat history retrieval; verified all API endpoints using **Postman**.
- Tech Stack: Node.js, Express.js, React.js, TypeScript, MongoDB, Socket.IO, JWT, RBAC, REST APIs, Postman*

Team Task Manager – Workflow and Project Management System | [GitHub](#)

2026

- Developed a full-stack **role-based task management system** supporting Admin and Member roles, with features for task assignment, lifecycle tracking, priority management, and delivery status monitoring across structured team workflows.
- Implemented **dashboard-based reporting views** for real-time team performance visibility and task ownership tracking; secured all API endpoints using **JWT and bcrypt** authentication; validated end-to-end workflows using **Postman**.
- Designed and optimized **MySQL (RDBMS)** schemas with targeted **SQL queries** to support audit logs and reporting pipelines; leveraged **MongoDB** for flexible document storage alongside structured relational data.
- Tech Stack: Node.js, Express.js, React.js, MySQL, MongoDB, JWT, REST APIs, SQL, Postman*

RESEARCH & LEADERSHIP

Research Publication · IEEE Xplore

2025

- Published **“Robotic Process Automation Tool to Enhance Efficiency in Public Services”** in IEEE Xplore, focusing on automation, process efficiency, workflow optimization, and technology-enabled service improvement.

Active Member & Volunteer · IEEE Student Chapter, Chandigarh University

Mar 2024 – Dec 2024

- Coordinated technical workshops and student events by collaborating with cross-functional teams, managing communication, tracking participation, and supporting execution planning.

INDUSTRIAL TRAINING

Social Internship Program — Chandigarh University, Mohali

Jun 2023 – Jul 2023

- Analyzed IoT energy datasets using **Python (NumPy, SciPy)** and **SQL**, improving data accuracy by **25%** through validation and root cause analysis in an **Agile/Scrum** team environment.

Blockchain Training Program — Metacrafters via Chandigarh University

Jun 2024 – Aug 2024

- Built and deployed **5+ smart contracts** on Ethereum and Polygon using **Solidity**, integrated with **Web3.js**, and configured basic **CI/CD** workflows for testing and deployment.

CERTIFICATIONS

Cloud Computing – Elite (Top 1%) — NPTEL, IIT Kharagpur

2024

Introduction to Networking and Cloud Computing — Microsoft via Coursera

2025

Foundations of Project Management — Google via Coursera

May 2026